

A CONDITIONAL SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON CANDIDATE, PENDING COEFFICIENT-TRACE COMPARISON

ANONYMOUS REPOSITORY AUDIT

ABSTRACT. We study a proposed trace-73 Cappell–Shaneson construction associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A public finite calculation produces the nonzero cubic Burau scalar -59072 in the grading convention corresponding to quantum degree 494 . An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid. We construct a public Aitchison–Rubinstein product-annulus presentation with an actual embedded detector collar. If the Burau functional is identified with the specified rational $N = 2$ Manolescu–Walker–Wedrich coefficient trace in the required symmetric-monoidal form, then the candidate is a homotopy sphere not diffeomorphic to S^4 . A relative complete-sphere argument shows that the three-handle closure then follows without an independent sphere-movie hypothesis, and published four-handle and standard-sphere results give the final identifications. The coefficient-trace identification remains open. The Lean development checks the finite and quotient algebra; it does not construct the external geometry or formalize four-dimensional differential topology.

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1. INTRODUCTION

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4 . Freedman’s work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

- (1) the candidate is a homotopy 4-sphere;
- (2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [12], and Iwaki [11]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [15]. Manolescu and Neithalath gave a two-handlebody description [13], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [14]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4 , which is concentrated in bidegree zero.

We investigate whether the public finite class can be constructed for the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. Compact product-ribbon, point-push, owner-lift and Nielsen ledgers replace parts of the unavailable historical data. The public AR product witness supplies the embedded framed Kirby presentation and detector collar, but not the candidate-specific quantum-trace map. Once that map is supplied in a relative monoidal form, a standard sphere system disjoint from the detector gives the three-handle closure. The paper therefore separates the conditional theorem from the remaining comparison gap.

Contributions. The main contributions are as follows.

- (1) We give a public parameterized Aitchison–Rubinstein product-annulus witness, two geometric cancellations, and an embedded six-sweep detector collar, thereby replacing the unavailable historical PD.
- (2) We identify the local checked R-matrix with the public Burau blocks and prove a full-entry third-order statement, while isolating the missing candidate-specific MWW-to-endpoint comparison and simultaneous transport.
- (3) We use the relative uniqueness of complete sphere systems and MWW’s intrinsic local module action to derive the three-handle closure from P0 and the symmetric-monoidal form of C, without TH-sized sphere movies.
- (4) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core.

Historical files referenced by earlier versions of the project are not publicly available. Their hashes are retained as provenance. The public AR witness replaces their P0 role; it does not yet supply the load-bearing MWW coefficient comparison C.

2. THE TRACE-73 CAPPELL–SHANESON CANDIDATE

Let

$$(1) \quad A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

This is Iwaki’s standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [11, Table 2]. Direct integer arithmetic gives

$$(2) \quad \det A = 1, \quad \det(A - I) = 1.$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \rightarrow T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε , where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki’s Proposition 2.1 and the original sources cited there [11, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing hypothesis below.

Iwaki proves standardness for many infinite families. The appearance of (41, 189, 73) in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada’s work stops below 73 [12]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. PRECISE STATEMENTS

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4 , and predicates $\text{HS}(M)$ and $\text{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem). *Suppose there are rational vector spaces W_0, W_1, W_2, W_3 , an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \quad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \quad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$(3) \quad \ell_0(x_0) = -59072,$$

$$(4) \quad \ell_1 q_{01} = \ell_0,$$

$$(5) \quad \ell_2 q_{12} = \ell_1.$$

Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \rightarrow S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then X is not diffeomorphic to S^4 .

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $-59072 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4 , the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. $\square$

The Lean theorem corresponding to Theorem 3.1 accepts the input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants of these structures are constructed in the repository. We name the candidate-specific assumptions needed to instantiate them.

Theorem 3.2 (Public AR presentation, P0). *The public AR product witness is a labelled and framed handle presentation of Σ_A^0 . Two product cancellations give the registered genus-two attaching link. A specified boundary 3-ball B*

contains an actual six-sweep detector collar inducing the public 44- and 88-strand braids on all oriented cable states. The presentation has three 3-handles followed by one final 4-handle.

Hypothesis 3.3 (Coefficient comparison, C). The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from B . Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is -59072 up to an invertible convention unit, and hence is nonzero in quantum degree 494.

Theorem 3.4 (Relative three-handle consequence, S). *Under Theorem 3.2 and Hypothesis 3.3, the three 3-handles may be attached along a complete sphere system disjoint from B . For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.*

Theorem 3.5 (Final identifications, P3). *MWW's three-/four-handle formulas identify the final quotient with $\mathcal{S}_{0,(0,494),0}^2(\Sigma_A^0; \mathbb{Q})$. The corresponding summand for S^4 is zero, and diffeomorphisms induce grading-preserving equivalences. Moreover, $\det(A - I) = 1$ makes Σ_A^0 a homotopy sphere.*

Theorem 3.6 (Conditional trace-73 theorem). *Assume Hypothesis 3.3. Then the resulting trace-73 candidate satisfies*

$$\Sigma_A^0 \simeq S^4 \quad \text{and} \quad \Sigma_A^0 \not\cong_{\text{diff}} S^4.$$

Proof of Theorem 3.6. Theorems 3.2 and 3.5, and the Cappell–Shaneson criterion give the homotopy-sphere assertion. Hypothesis 3.3 supplies the coefficient comparison, and Theorem 3.4 supplies the three-handle descent. They instantiate Theorem 3.1, whose selected class is nonzero in quantum degree 494. MWW's four-handle isomorphism preserves this degree, while the same degree of the standard S^4 module vanishes. Diffeomorphism invariance and Theorem 3.1 rule out a diffeomorphism to S^4 . $\square$

4. THE FINITE COMPUTATIONAL LAYER

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

4.1. Integral lattice calculations. Besides (2), the repository records the proposed sphere coordinate columns

$$(6) \quad C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \quad \det C = 1.$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\text{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3; \mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincaré duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

4.2. The public Burau computation. Let $\varepsilon = t - 1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

- (1) parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
- (2) cable it to a 45,360-letter word on 88 strands;
- (3) apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
- (4) substitute $t = q^{-2}$ with $q = 1 + h$, equivalently $\varepsilon = (1 + h)^{-2} - 1$;
- (5) extract the coefficient of h^3 after the specified source vector and covector are applied.

Denote the resulting scalar by δ_3^{Bur} . The independently reproduced public result is

$$(7) \quad \delta_3^{\text{Bur}} = -59072 = (-2)^3 \cdot 7384.$$

The public input does not contain either decimal literal 7384 or -59072 ; the former is, however, stored as the constant `cubicBase` in the Lean finite layer. Lean checks only the arithmetic $(-2)^3 \cdot 7384 = -59072$ and never evaluates the 45,360-letter word.

Proposition 4.1 (Exact scope of the computation). *Equation (7) is a statement about the specified truncated Burau model. The script and JSON alone do not define the MWW invariant. Under the coefficient-trace comparison of Theorem 3.3, however,*

$$\delta_3^{\text{MWW}} = \delta_3^{\text{Bur}} = -59072,$$

up to the harmless global convention unit normalized in that theorem.

Proof. The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. The displayed MWW equality is exactly the additional content of Hypothesis 3.3; it does not follow from the finite script or from the local Burau-block calculation. $\square$

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

5. LASAGNA MODULES AND THE INTENDED OBSTRUCTION

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W , modulo local relations coming from KhR_N cobordism maps [15, 14].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a co-equalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to S^4 . The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections record partial constructions and the exact candidate-specific inputs still needed by that algebra.

6. COMPACT PRODUCT-RIBBON PRESENTATION

We prove Theorem 3.2. The construction below replaces, rather than reconstructs, the unavailable historical planar diagram.

6.1. The actual AR link. Use the coordinate-spine genus-three Heegaard splitting of T^3 from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a [public Berkeley scan](#). Let C_i be the three coordinate spine circles with common base vertex Q , and let A_i be pairwise-disjoint narrow disk neighborhoods away from Q . Write t for the mapping-torus one-handle. If ψ_A is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the i th mapping-torus two-handle is the embedded circle

$$(8) \quad C_i^- \cup \lambda_i \cup \psi_A(C_i)^+ \cup \mu_i,$$

where λ_i, μ_i are the two arcs through t . The framing annulus is constructed simultaneously as

$$(9) \quad A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I).$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (8)–(9) define the whole embedded framed link, not merely its words.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \quad r_{yz} = [y, z], \quad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle h_{CS} goes geometrically once over t . Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in indices $0, \dots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \quad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \quad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by R transports the complete AR construction to the displayed matrix A . Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

6.2. Return to the linear torus map. Let ϕ_A be the linear torus map, straightened to the identity on the section ball, and choose an isotopy ρ_u , $u \in [-1, 1]$, from ϕ_A to ψ_A , stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$(10) \quad F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A}$$

defines a diffeomorphism. Indeed, $g_{-1} = \text{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (10) smooth, and the family g_u^{-1} gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by F replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

6.3. Words and normal fields. Lift C_i and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to Ae_i ; it crosses the j th integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$(11) \quad m_i = t\phi_A(x_i)t^{-1}x_i^{-1}.$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1, 1, 1)$; the resulting strips have parallel boundary components, and linearity of A preserves this property on their images [1, pp. 16–17]. Together with the λ_i/μ_i rectangles, these strips are the normal fields in (9). Thus no blackboard integer is substituted for a framing.

6.4. The two cancellations. Aitchison–Rubinstein identify (t, h_{CS}) as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over h_{CS} along its product rectangle and cancel; this removes the t, t^{-1} passages with zero relative twist.

Now $Ae_1 = e_3$, so the actual curve m_1 becomes zx^{-1} and has one geometric passage over the x one-handle. Thus (x, m_1) is another complementary pair. Sliding every other x passage over m_1 along parallel product bands replaces x by z and cancels the pair. All labels and normal fields move with the bands. The resulting counts are $(1, 2, 5, 3, 1)$, and the exact word projection is

$$\begin{aligned} |m_2| &= 311, & [m_2]_{(y,z)} &= (40, 269), \\ |m_3| &= 1460, & [m_3]_{(y,z)} &= (189, 1271). \end{aligned}$$

The transported r_{zx} has empty free word, but no split-unknot conclusion is needed or inferred from that fact.

6.5. The actual detector ball. Cut the remaining genus-two handlebody along its y, z disk system. The selected m_2/r_{xy} state has 42 y passages from m_2 and two from r_{xy} . At distinct product-normal levels, a regular neighborhood of these 44 passages is a standard three-ball B containing a punctured-disk collar. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of B . Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy

simultaneously. The public verifier regenerates all 252 factors and obtains

$$\begin{aligned} |B_{44}| &= 11340, \\ \text{SHA256}(B_{44}) &= \text{7C2D2F792C2672221A76CAF08A71F560A} \\ &\quad \text{FOCB7654B4D537C00FAD00B16EFA187} \quad . \\ \text{perm}(B_{44}) &= 1, \quad \text{wr}(B_{44}) = 0. \end{aligned}$$

The exact smooth construction, source digest, matrix bridge, handle counts, cancellation data and collar labels are regenerated in `audit/t73_ar_product_witness.json`. Its digest is

$$\begin{aligned} &\text{AA9328EA3C7A53B937C97A02D611850D} \\ &\text{0E8EEAF87C68756C04B013A486F6F112.} \end{aligned}$$

The source-PDF, regeneration and mutation checks are described in the repository README. This proves Theorem 3.2 without identifying the retired historical PD.

7. THE COEFFICIENT-TRACE COMPARISON

The compact words give

$$p_y = 42 + 2 = 44, \quad p_z = 269 + 2 = 271.$$

The counts suggest product subrectangles pairing the 44 y wickets with 44 z passages and 227 remaining circles. The desired cut coefficient would then have the two-sided Hattori form

$$M_R(T, T') \cong \text{Hom}(B_{act}T, B_{act}T') \{-44\} \otimes A^{\otimes 227}, \quad A = \mathbb{Q}[X]/(X^2).$$

MWW Theorem 4.7 identifies the one-handle module with the coefficient trace of the cut tangle categories and describes cobordism maps component-wise [14]. BPW’s functorial quantum vertical-to-horizontal trace and annular functor give

$$q \text{Tr}(\mathcal{C}; M_R) \longrightarrow h \text{Tr}_q(BN) \xrightarrow{F_{Aq}} \text{gRep}(U_q(\mathfrak{sl}_2))$$

as constructed by BPW [4]. BHPW supplies strict integral tangle/foam functoriality [3]. The base change $q \mapsto 1 + h$ is a rational localization followed by a Noetherian adic completion and is flat.

The endpoint type must be stated carefully. Put

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \quad E_{88} = (V^{\otimes 88})_{\lambda=86}.$$

Then E_{86} has rank one and E_{88} has rank 88; after restriction to weight spaces the cup lies in $\text{Hom}_{\mathbb{Q}[[h]]}(E_{86}, E_{88}) \cong E_{88}$. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before this restriction. It is therefore incorrect to place a “weight-86 summand” on an intertwiner space itself.

There is also no abstract obstruction at the representable-coefficient step. For a rational-linear autofunctor B , the coefficient bimodule $(T, T') \mapsto \text{Hom}(BT, BT')$ is two-sided equivalent to the regular coefficient bimodule, and hence their coefficient HH_0 groups are linearly equivalent. This quotient statement is kernel checked in `Smooth4PC/RepresentableCoefficient`.

lean. Its use here still requires the missing candidate-specific, action-compatible Hattori equivalence from M_R to that representable model, as well as its q -deformed completed lift.

On the weight-one subspace, the normalized checked R-matrix is

$$\begin{pmatrix} 1 - q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After conjugation by $\text{diag}(1, q^{-1})$ and setting $t = q^{-2}$, this becomes the public Burau block. This is a valid local comparison. It does not construct a map from the candidate's MWW coefficient trace to the specified qHH0/Chern summand, nor the single coordinate change P that must transport the actual operator, source vector, and covector simultaneously.

The compact word has zero writhe and identity permutation. Direct evaluation of all 88 basis vectors proves

$$\rho_h(W) - I \in h^3 \text{End}(E_{88}).$$

An exact noncommutative Magnus expansion of the Artin action on F_{44} shows that the automorphism is the identity through degrees one, two and three. All intermediate integer coefficients have absolute value at most four. Darné's Andreadakis equality for the pure braid group then gives

$$W \in \Gamma_3(P_{44}).$$

See [6, Theorem 6.2]. Every physical cabling is a group homomorphism and hence preserves the lower central series. Since each local pure generator acts as $I + O(h)$, every statewise cabling satisfies $\rho_h(W_s) - I = O(h^3)$. If the missing candidate-specific map and simultaneous transport exist, BPW's tangle action gives statewise equations $Q_e W_s = W_t Q_e$ for the beta and psi edges. General functoriality transports an identified action; it does not construct this missing arrow. Finally, the core identities

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \quad (\epsilon \otimes \epsilon)\Delta(X) = 1$$

give the required Frobenius equations after the actual maps have been bound. Thus the public results prove the order-three and local algebraic parts of C, while the coefficient-trace comparison and complete beta/psi descent remain Hypothesis 3.3.

8. THREE-HANDLE SPHERE CLOSURE

We prove Theorem 3.4. Remove the final 4-handle and call the remaining handlebody W_3 . Its boundary is S^3 . Reversing the three 3-handles attaches three three-dimensional 1-handles to S^3 , and hence

$$\partial W_2 \cong \#^3(S^1 \times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of W_3 .

Fix the detector ball B from P0. Choose a small ball in the connected complement of the actual attaching system. Since any two orientation-preserving

embedded 3-balls in a connected oriented 3-manifold are ambient isotopic, the actual system may first be isotoped off the fixed ball B . Choose the standard three nonseparating spheres in the complement of B . Their complement is connected and their classes form the standard basis of $H_2^{\text{sph}}(\partial W_2)$. Horvat–Jabłonowski’s Theorem 5.3 identifies this system with the actual attaching system up to permutation, 3–3 handle slides and isotopy [10]. Applying their relative uniqueness lemma to $Q = \partial W_2 \setminus \text{Int } B$ makes the isotopies fixed on ∂B . The additional boundary slides become tubings to the inessential sphere ∂B after the ball is restored and are removed by a local isotopy. Thus the 3-handles may be represented by standard spheres without moving the detector.

MWW’s intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let A_0 denote the essential sphere with one dot and A_1 the undotted sphere. The quotient relations are

$$A_0 = 1, \quad A_1 = 0.$$

The symmetric-monoidal naturality required in C applies because the standard spheres lie outside B . For every element v in the complete source,

$$\ell(vA_0) = \ell(v) \text{ev}(\dot{S}^2) = \ell(v), \quad \ell(vA_1) = \ell(v) \text{ev}(S^2) = 0.$$

These are whole-source equations. Hence the detector annihilates the full relation submodule for each of the three standard spheres and descends through the iterated MWW coequalizer. This proves Theorem 3.4 from P0 and C.

The owner lifts, the 32-step Nielsen ledger and the split-tree Lean lemmas are compatible optional coordinate models, but they are not needed for this relative proof. Candidate-level S is therefore not an independent open input; it remains unavailable only because P0 and the full monoidal form of C are not yet instantiated.

9. FINAL IDENTIFICATIONS

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-(0,0) isomorphism [14]. By Theorem 3.2, this is the handle decomposition of Σ_A^0 . If C and Theorem 3.4 have produced the specified selected class, it remains nonzero in

$$\mathcal{S}_{0,(0,494),0}^2(\Sigma_A^0; \mathbb{Q}).$$

MWW Corollary 3.5 gives $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$, concentrated in bidegree (0,0), so the corresponding rational degree-494 summand vanishes. The intrinsic definition transports lasagna fillings under diffeomorphisms and gives grading-preserving equivalences. Finally, Iwaki’s Proposition 2.1 and (2) prove that the genuine Cappell–Shaneson manifold Σ_A^0 is a homotopy sphere [11]. Applying these general results to the identified handle decomposition proves

Theorem 3.5. Hypothesis C is needed only to supply the nonzero degree-494 class to which these identifications are applied.

10. FORMALIZATION BOUNDARY

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

$$\text{ExternalGeometry} \longrightarrow \text{CSExternalGeometry} \longrightarrow (\text{HS}(X) \wedge \neg \text{Diff}(X, S^4)).$$

It does not construct either input structure.

The finite files check:

- the matrix A , $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;
- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

$$\begin{aligned} &\text{IsActualHHO}, \quad \text{IsActualSphereMap}, \\ &\text{IsActualMWWCoequalizer}, \quad \text{IsActualFourHandle}. \end{aligned}$$

These are predicate fields of the ambient universe rather than definitions of smooth topology. The present paper does not supply their mathematical instances: C still states missing candidate-level data, while the P0 and S fields are supplied mathematically outside Lean by Theorem 3.2 and its relative three-handle consequence. It would be incorrect to call the Lean development a formalization of four-dimensional differential topology or an unconditional proof.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core. P0, S and P3 are established mathematically in Sections 6, 8 and 9, not by the Lean kernel; the coefficient comparison C remains external and open.

11. STATUS TABLE AND REPRODUCIBILITY LEDGER

The table records the current allocation of every external layer. DISCHARGED means that a public generator or primary theorem and an independently checkable argument are both supplied.

Item	Verdict	Reason
P0	DISCHARGED	The public AR source, parameterized product-annulus witness, two geometric cancellations and embedded detector ball identify the replacement presentation with Σ_A^0 .
P1/C	OPEN	Cubic order and the ordinary representable-coefficient reduction are proved, but the actual product bimodule equivalence, its completed lift and simultaneous transport are missing.
P2/E7	DISCHARGED	P0 supplies the handle pattern and detector ball; HJ and the relative complete-system lemma give standard attaching spheres disjoint from the detector.
P2/E10/S	PARTIAL	MWW's intrinsic local module action and monoidal C imply the complete sphere relations; no independent sphere movie remains, but C is open.
P3/E11	DISCHARGED	P0 fixes the actual handle decomposition and MWW Proposition 3.4 gives the bidegree-(0,0) four-handle isomorphism.
P3/E12	DISCHARGED	MWW Corollary 3.5 gives the standard S^4 module concentrated in bidegree zero.
P3/E13	DISCHARGED	Theorem 3.2 identifies Σ_A^0 and Iwaki's determinant criterion applies.

The historical availability script reports the large files as absent. The compact generators provide useful partial replacements but not the embedded geometry and functorial maps listed in the table.

12. RELATED WORK

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 12, 11]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for (41, 189, 73) is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [13, 14]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]; the candidate-specific comparison with the public endpoint coordinates remains open.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [16]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [10]. It reduces the replacement problem to boundary, embeddedness, disjointness and homology-basis checks. Applied relative to the detector ball, its complete-system lemma supplies the

standard attaching system used here; the owner-lift and Nielsen ledgers are optional coordinates.

13. REPRODUCIBILITY AND REVIEW

The proved finite and conditional layers can be replayed in the following order.

- (1) Regenerate the public AR product witness, point-push, Hattori and owner-lift ledgers.
- (2) Verify the full 88-dimensional order-three matrix statement and its mutation controls.
- (3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
- (4) Audit Hypothesis C and the relative-S consequence against the cited BPW/BHPW/MWW functoriality and handle formulas.
- (5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.

The historical large artifacts are unavailable for this replay. Independent review should check the public AR mapping-torus diffeomorphism, product framings and two cancellations, then concentrate on the proposed BPW coefficient comparison in Hypothesis C and its monoidality used by the relative sphere theorem.

14. CONCLUSION

The finite calculation and exact Magnus certificate provide a reproducible cubic-order signal for the public braid, and the Lean development verifies the abstract nonvanishing chain. A counterexample conclusion would follow from Theorem 3.6, but C remains open: the present data do not identify a genuine monoidal MWW coefficient functional with the public Burau detector. P0 is supplied by the public AR product witness, and the relative sphere theorem makes S a consequence of C rather than an independent hypothesis. The paper therefore establishes a conditional theorem and a precise closure programme, not a disproof of the smooth four-dimensional Poincare conjecture.

APPENDIX A. RETIRED HISTORICAL OBJECTS

The following table records historical artifacts cited by earlier project notes. They are unavailable; the compact partial constructions do not yet replace the P1 coefficient-comparison role. The AR and relative-sphere replacements discharge the old P0/P2 roles. The digests are retained for provenance.

Priority	Object	SHA-256
P0	cs_presentation.py	B20BDF6FBF2A0E629083E2E62DD1DDF 9172A4E8893311CE122FFC4FEB388DFA 6DF4D7906ADB18BA2D3237D4B49F8985
P0	t73_eps0.erkmo.json	A5B18D8C39FF522AD6498F2369803FCB E6912A64457557469E5C691B4D57ABDB BF4C45ADB05492777C574223D0C06F8A
P0	t73_reduced_billiard.pd.json	4E1C4564ED98F30EAEBOA1AEAAD95ACA B5AD7B226C79B7D5BF3F259E8B56F541 2D9E4760F836D9E04E6E0024CAE03A3D
P0	CUT_OBJECT.json	BB49A74FF5D129756F25450B30F10F7C B2E97EB0981036768B8F1C1D5EF16F8 817EC1A7B951B6B32DBA2F9211FFD725A
P0	NORMAL_FIELD_MOVIE_CERT.json	13AED414447ED33057C99D340910B501 09983CF36874C14ECE0E6A2834A12C24 9D93A6CB320200A906F461EF3131127E
P0	verify_t73_collar_braid.py	A40C80A9A7828128E6F2804808417056 7F3D3618D6A790A9B60EE8085B647AC2 AB742E1BC9C15841F1BEF015034217B5
P1	ACTUAL_PD_CABLE_UNIT_CERT.json	DE1AC479699EC79DE76D4265993DE437 493A3AAA6CABB636F98998644BF3181C EE620E6B085A5F9E1C73CFDD1AD04FC
P1	PRODUCT_NORMAL_CHRISTOFFEL_THXY_MOVIE.json	0682CEC74DA3DBF8AFE70DD19C038E3A0 4D1B627C0343A1C464319704EAFADCA1 27902A2E4E90CF1C283004359B7ADC24
P2	TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json	EABF67C0D1AE309F0281297710A283B8 ED5A11D88C8E810837932313F4C53227
P2	TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json	
P2	THXY_FULL_MACRO_P3FREE_HJ_CERT.json	

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. These objects are not included. The public AR replacement discharges P0 without them. A proof of C must still recover or replace the actual coefficient/Hattori object and its functorial role. The historical P2 objects are no longer required by the relative sphere argument.

APPENDIX B. CORRESPONDENCE WITH THE LEAN FIELDS

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
x0, ell0, ell0_x0	P1 comparison and the selected genuine MWW class.
q01, ell1_comp_q01	Complete two-handle beta/psi quotient and descent.
q12, ell2_comp_q12	Actual E7 spheres, MWW hemisphere maps, and complete coequalizer.
transport, fourIso	E11 applied to the same graded lasagna module.
s4DegreeZero	E12 after identifying G with rational $N = 2$ lasagna.
diffeomorphismEquiv	Graded diffeomorphism invariance of that same object.
matrixConditionsToHomotopySphere	E13 for the manifold identified by P0.

These are exactly the candidate-level obligations isolated in Sections 5–8. They have not been discharged. The Lean theorem consumes no stronger statement, but it still requires them as structure fields.

REFERENCES

1. Iain R. Aitchison and J. Hyam Rubinstein, *Fibered knots and involutions on homotopy spheres*, Four-Manifold Theory (Durham, N.H., 1982) (Cameron McA. Gordon and Robion C. Kirby, eds.), Contemporary Mathematics, vol. 35, American Mathematical Society, Providence, RI, 1984, pp. 1–74. MR 780575
2. Selman Akbulut, *Cappell–shaneson homotopy spheres are standard*, Annals of Mathematics **171** (2010), no. 3, 2171–2175.
3. Anna Beliakova, Matthew Hogancamp, Krzysztof Karol Putyra, and Stephan Martin Wehrli, *On the functoriality of $\mathfrak{sl}_2$ tangle homology*, Algebraic & Geometric Topology **23** (2023), no. 3, 1303–1361.
4. Anna Beliakova, Krzysztof Karol Putyra, and Stephan Martin Wehrli, *Quantum link homology via trace functor I*, Inventiones Mathematicae **215** (2019), no. 2, 383–492.
5. Sylvain E. Cappell and Julius L. Shaneson, *Some new four-manifolds*, Annals of Mathematics **104** (1976), no. 1, 61–72.
6. Jacques Darné, *On the andreadakis problem for subgroups of ia_n* , International Mathematics Research Notices **2021** (2021), no. 19, 14720–14742.
7. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
8. Michael Hartley Freedman, *The topology of four-dimensional manifolds*, Journal of Differential Geometry **17** (1982), no. 3, 357–453.
9. Robert E. Gompf, *More cappell–shaneson spheres are standard*, Algebraic & Geometric Topology **10** (2010), no. 3, 1665–1681.
10. Eva Horvat and Michał Jabłonowski, *On 4-dimensional 3-handle attachments*, 2025.
11. Kazunori Iwaki, *Infinite families of standard cappell–shaneson spheres*, 2024.
12. Min Hoon Kim and Shohei Yamada, *Ideal classes and cappell–shaneson homotopy 4-spheres*, 2017.
13. Ciprian Manolescu and Ikshu Neithalath, *Skein lasagna modules for 2-handlebodies*, 2020.
14. Ciprian Manolescu, Kevin Walker, and Paul Wedrich, *Skein lasagna modules and handle decompositions*, Advances in Mathematics **425** (2023), 109071.
15. Scott Morrison, Kevin Walker, and Paul Wedrich, *Invariants of 4-manifolds from khovanov–rozansky link homology*, 2019.
16. Qiuyu Ren and Michael Willis, *Khovanov homology and exotic 4-manifolds*, 2024.

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